AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING

Postgraduate Programs
Spring 2026

CSE323: Advanced Embedded Systems Design
CSE411/CSE411s: Real-Time and Embedded Systems Design

Assignment-05

Objective

The goal of this assignment is to understand the behavior and limitations of Binary Semaphores in FreeRTOS, specifically regarding how they handle multiple pending interrupts.

Description

You are required to develop a FreeRTOS application that demonstrates a scenario where a Binary Semaphore is used for synchronization between an Interrupt Service Routine (ISR) and a handler task.

Specifically, you must simulate a condition where the number of pending interrupts occurring during the task's execution is greater than one, and demonstrate how the system responds to this overload.

Task Requirements

Your implementation must clearly demonstrate the following behavior:

- **Trigger Multiple Interrupts:** Simulate a scenario where 2 or 3 hardware interrupts fire in quick succession while the handler task is busy executing and has not yet completed its current cycle.
- **Binary Semaphore Behavior:** Use a Binary Semaphore to signal the task from the ISR (xSemaphoreGiveFromISR).
- **Observation of Dropped Events:** Prove that because a Binary Semaphore can only hold a maximum count of 1, only **one** of those pending interrupts is ultimately serviced by the task once it resumes.
- **Result:** Clearly show that the remaining interrupts (1 or 2) are entirely lost or dropped.

What your report must demonstrate

- A clear explanation of your system setup (which interrupt source you used and the execution times of your tasks).
- Supporting technical evidence that explicitly proves multiple interrupts fired, but only one was serviced by the task.

Submission format

Submit **ONE PDF document only**, containing your setup, methodology, and the screenshot evidence proving the lost interrupts.